

Proof skeleton

Exact Diffusion Design for Maximally Collective Stable Turing Patterns in Binary-Complex Mass-Action Networks

Topology-Wide All-Spectrum Localization and Exponent-Optimal Stationary Heterogeneity Trade-Offs

1. Reaction algebra and topology-wide localization

Construct Y_m and Γ_m directly from the indexed reactions. Reaction-by-reaction verification gives $c^T \Gamma_m = 0$ for $c = (0, 4, \dots, 4, 2, 1)^T$. Solving the balance equations gives exactly

$$\ker \Gamma_m = \text{span}\{(\mathbf{1}_m, 0, 0), (0_m, 1, 1)\}.$$

Because Γ_m has $m + 2$ columns, rank-nullity gives $\text{rank } \Gamma_m = m$. Independently, the maximal minor on rows $X_1, X_3, \dots, X_m, Z$ and columns $R_2, \dots, R_{m-2}, R_a, R_b, R_+$ has determinant $4(-1)^m$. Therefore every positive flux is $(a\mathbf{1}_m, b, b)$. Differentiation at the unit equilibrium gives $A_m(a, b)$; at a general equilibrium $x^* = H^{-1}\mathbf{1}$, the Jacobian is $A_m(a, b)H$. Conversely, the choice

$$k_r = \frac{v_r}{(x^*)^{y_r}}$$

realizes every $a, b, H > 0$.

For a nonempty principal set with fewer than m vertices, Frobenius decomposition reduces every nontrivial SCC to: the $X_1, \dots, X_{m-1}$ long cycle; the $X_2, \dots, X_m$ long cycle; a principal boundary-triad block; or a negative singleton. At $b = 2a$ one directed edge disappears and only refines the decomposition. The two cycle characteristic equations are excluded from $\Re \lambda \geq 0$ by strict product inequalities. The full triad has positive cubic coefficients and $c_1 c_2 - c_3$ is a sum of fourteen positive monomials. Every lower-order principal matrix is thus Hurwitz. Sparse expansion of the $X_1, \dots, X_m$ block gives the negative signed determinant $-2a^{m-1}b \prod h_i$, forcing a positive real eigenvalue.

For $m = 3$, the two long cycles are $\{X_1, X_2\}$ and $\{X_2, X_3\}$, and every other nontrivial SCC is a principal subgraph of $\{X_1, X_3, Z\}$. Thus the chain argument is applied only for $m \geq 4$.

Audit object. Independently generate every induced SCC for $m = 3, \dots, 8$ at generic b and at $b = 2a$, while the proof uses the all- m graph classification rather than enumeration.

2. Principal-minor diffusion-ray theorem and exact network law

For $a_I = (-1)^{|I|} \det J_{I,I}$ and $D \succ 0$, multilinearity gives

$$\det(sD - J) = \sum_I a_I \prod_{j \notin I} s d_j = s(\beta_1 + \beta_2 s + \cdots + \beta_n s^{n-1}).$$

Assume directly that $a_I > 0$ for $|I| \leq n-2$ and that $\sum_{|I|=n-1} a_I > 0$; a simple zero and otherwise stable spectrum supplies the latter coefficient condition in the application. Hence $\beta_k > 0$ for $k \geq 2$, and the bracket is strictly increasing on $s > 0$, proving the unique simple threshold iff $\beta_1 < 0$.

For the stronger positive-real band, define

$$\chi_s(\lambda) = \det(\lambda I + sD - J) = \sum_I a_I \prod_{j \notin I} (\lambda + s d_j).$$

Every derivative contribution from $|I| \leq n-2$ is nonnegative. The total order- $(n-1)$ derivative is $\sum_{|I|=n-1} a_I > 0$, the coefficient of λ in $\det(\lambda I - J)$. Hence $\chi'_s(\lambda) > 0$ for $\lambda \geq 0$. The sign of $\chi_s(0)$ therefore gives a positive real eigenvalue exactly for $0 < s < s_*$. At threshold, $\chi_{s_*}(0) = 0$ and $\chi'_{s_*}(0) > 0$; this, rather than scalar simplicity of the generalized-pencil root alone, proves ordinary algebraic simplicity of zero for $J - s_* D$.

For $A_m(a, b)H$, compute every order- m omission minor. Omitting Z gives weight -2 ; omitting an interior X_j gives $+16$; omitting X_1 or X_m gives zero. Thus

$$\beta_1 = 2a^{m-1}b \prod_{i=1}^m h_i \left(8h_Z \sum_{j=2}^{m-1} \frac{d_j}{h_j} - d_Z \right).$$

This proves the exact criterion. The same sign follows from first-order perturbation of the conservation eigenvalue. For fixed H , minimizing $d_{\max}/d_{\min}$ under the strict inequality gives the nonattained infimum $T(H) = 8h_Z \sum_{j=2}^{m-1} h_j^{-1}$. At $H = I$, $T = 8(m-2)$, and $h_Z/h_j \geq 1/\chi_H$ gives $\chi_D \chi_H > 8(m-2)$.

3. Improved unit-equilibrium primary bifurcation

Set $a = b = 1$, $H = I$, $K_i = 91m - 181 - i$, and use

$$d_1 = 23/63, \quad d_i = K_i^{-1}, \quad d_m = 1/7, \quad d_Z = 16/45.$$

An explicit right vector satisfies $(A_m - D_m)r = 0$ and an explicit left vector satisfies $(A_m - D_m)^T \ell = 0$. Exact harmonic-sum formulas prove $\ell^T r < 0$ and $\ell^T D_m r < 0$, hence transversality $\eta_m > 0$.

Homogeneous stability follows from

$$\det(\lambda I - A_m) = \lambda q_m(\lambda), \quad \lambda q_m = (1 + \lambda)^{m-3} P(\lambda) - R(\lambda),$$

where $P = \lambda^4 + 12\lambda^3 + 42\lambda^2 + 47\lambda + 16$ and $R = 5\lambda^2 + 33\lambda + 16$. The exact difference $|1 + \lambda|^2 |P|^2 - |R|^2$ expands into 35 nonnegative monomials in $x = \Re \lambda$ and $y^2 = (\Im \lambda)^2$, with equality only at the origin. For $m = 3$, $q_3 = (\lambda + 7)(\lambda^2 + 5\lambda + 2)$ is Hurwitz. For $m \geq 4$, $|1 + \lambda| \geq 1$ makes the 35-term strict inequality incompatible with $(1 + \lambda)^{m-3} P = R$ at any nonzero closed-right-half-plane root. Since $q_m(0) = 16m - 34 > 0$, the conservation zero is simple.

For spatial damping tD_m , set $g_1 = \lambda + 2 + (23/63)t$, $g_m = \lambda + 5 + t/7$, $g_Z = \lambda + 4 + (16/45)t$, $F = g_1 g_m g_Z - 4g_1 - 4g_m + g_Z$, and $G = g_Z(4g_1 + g_m) - 36$. Chain elimination gives $\det(\lambda I - A_m + tD_m) = Q_m F - G$. At $(\lambda, t) = (0, 1)$, the interior product telescopes to $91/90$. With $z = y^2$ and $s = t - 1$, the exact source

$$E_{77} = [(91/90)^2 + z] |F(x + i\sqrt{z}, 1 + s)|^2 - |G(x + i\sqrt{z}, 1 + s)|^2$$

has 77 strictly positive coefficients and no constant term. Thus equality occurs only at $(x, z, s) = (0, 0, 0)$. Hence the first mode has one simple zero, all other first-mode eigenvalues are stable, and all $t > 1$ modes are Hurwitz.

The exact unit contrast is

$$\chi_D = \frac{23(91m - 183)}{63},$$

which improves the superseded high-contrast profile and is within a dimension-independent factor of the sharp linear threshold.

4. Conservation-compatible normal form and stable branch

The mass-action field is quadratic:

$$f(\mathbf{1} + u) = A_m u + \frac{1}{2} B_m(u, u), \quad c^T B_m(u, v) = 0.$$

On the fixed integrated-mass class, the critical mode is $r \cos \xi$. Since $\cos^2 \xi = (1 + \cos 2\xi)/2$, solve

$$A_m w_0 = -\frac{1}{4} B(r, r), \quad c^T w_0 = 0, \quad (A_m - 4D_m)w_2 = -\frac{1}{4} B(r, r).$$

The first is an augmented nonsingular solve; the second is invertible by mode isolation. The interior w_2 recurrence telescopes through four consecutive K_i factors, leaving an explicit 4×4 system in the four retained boundary values. Its determinant is $64\mathcal{Q}_m/[6615(91m - 183)(91m - 181)(91m - 180)] > 0$.

The fixed-mass sectorial semiflow has a one-dimensional center manifold. The spatial reflection $\xi \mapsto \pi - \xi$ commutes with the Neumann semiflow, preserves the mass class, and sends $r \cos \xi$ to its negative. An equivariant center manifold therefore has an odd reduced vector field, and reflection exchanges the two nonzero branches. Projection of its reduced vector field onto the adjoint critical mode yields

$$\begin{aligned} \dot{A} &= \eta_m \mu A + c_m A^3 + O(|A|^5 + |A|^3 |\mu| + |A| |\mu|^2), \\ c_m &= \frac{\ell^T \{B(r, w_0) + \frac{1}{2} B(r, w_2)\}}{\ell^T r}. \end{aligned}$$

Setting this reduced vector field to zero gives the stationary Lyapunov–Schmidt equation. Writing $\mathfrak{h}_m = \sum_{j=1}^{m-2} K_j^{-1}$, the numerator telescopes to $R_m + C_m \mathfrak{h}_m$. The supplement prints R_m , C_m , their positive denominators, and the exact L_m clearing identity. Shifted coefficientwise-positive polynomials prove the numerator positive, while $\ell^T r < 0$; therefore $c_m < 0$ for every $m \geq 3$. For a direct audit, put

$$\mathcal{Q}_m = 589180301m^3 - 3500015940m^2 + 6930529579m - 4574434500,$$

$$\begin{aligned} P_R &= 68605040480814208768m^4 - 550882186169626030957m^3 + 1658612632937449670852m^2 \\ &\quad - 2219226476204103501323m + 1113379274975809565700, \\ P_C &= 652054120726848m^4 - 5151971981328467m^3 + 15265080924982572m^2 \\ &\quad - 20102347725659113m + 9927281930180400. \end{aligned}$$

Then

$$R_m = \frac{P_R}{286118780220(8m - 17)\mathcal{Q}_m}, \quad C_m = -\frac{215P_C}{11645046(8m - 17)\mathcal{Q}_m},$$

and exact clearing gives

$$R_m + C_m \frac{m - 2}{90m - 179} = \frac{L_m}{286118780220(8m - 17)(90m - 179)\mathcal{Q}_m}.$$

The shifted coefficients of $\mathcal{Q}_m, P_R, P_C, L_m$ are positive, so $C_m < 0$ and $R_m + C_m \mathfrak{h}_m \geq R_m + C_m(m - 2)/(90m - 179) > 0$.

Crandall–Rabinowitz gives the stationary local curve; reflection equivariance and the odd normal form identify its two positive nonconstant halves for $\mu > 0$. Their center eigenvalue is $-2\eta_m \mu + O(\mu^2) < 0$. The complementary spectrum is separated at onset and remains separated under analytic perturbation. Positive Neumann diffusion is sectorial on fixed-mass L^2 , its fractional H^1 domain is a Banach algebra in one dimension, and the quadratic Nemytskii map is smooth. The principle of linearized stability gives local exponential convergence in H^1 for sufficiently close nonnegative initial data in the same conservation class.

5. Equilibrium-scaled stable trade-off and audit boundaries

Let $\nu = m - 2$ and

$$\kappa_\nu = \begin{cases} 1/\sqrt{3}, & \nu = 1, \\ \sqrt{5}/2, & \nu \geq 2, \end{cases} \quad L_0 = \frac{\kappa_\nu}{\sqrt{\nu}}, \quad L_1 = \frac{90\nu}{90\nu + 1}.$$

The interval is nonempty: for $\nu = 1$ the squared inequality is $8281 < 24300$, while for $\nu \geq 2$ it follows from $(90\nu + 1)^2 \leq (91\nu)^2 < 6480\nu^3$. Set $h_1 = h_m = h_Z = 1$, $h_i = K_i/(LK_{i-1})$, and choose physical diffusion $D^{\text{phys}} = H_m(L)\Delta$. Under $\hat{x} = H_m(L)x$,

$$\partial_t \hat{x} = H_m(L)\{f(\hat{x}) + (1 - \mu)\Delta\partial_{xx}\hat{x}\}.$$

At $\mu = 0$, the effective stationary bracket and right vector are unchanged; the dynamic left vector is $\tilde{\ell}_m(L) = H_m(L)^{-1}\ell_m$. The physical fixed-mass functional is $(H_m(L)^{-1}c)^T \hat{x}$. Moreover,

$$\tilde{\ell}_m(L)^T H_m(L)\Delta r_m = \ell_m^T \Delta r_m < 0, \quad \tilde{\ell}_m(L)^T r_m < 0,$$

so the scaled crossing coefficient $\eta_m(L)$ is positive.

Chain elimination now gives interior factors

$$g_i = \frac{K_{i-1}}{K_i}(1 + L\lambda) + \frac{t-1}{K_i}.$$

For $t \geq 1$, Bernoulli's inequality and $\nu L^2 \geq 1/3$ reduce spatial half-plane exclusion to the exact source polynomial

$$E_{84} = (91/90)^2(1 + Ax + z/3)|F(x + i\sqrt{z}, 1 + s)|^2 - |G(x + i\sqrt{z}, 1 + s)|^2, \quad A = 2\nu L, z = y^2, s = t - 1.$$

Its 84 coefficients are coefficientwise nonnegative in $A > 0$, and equality occurs only at the first-mode zero $(x, z, s) = (0, 0, 0)$.

At $t = 0$, the exact homogeneous determinant, after dividing by $\det H_m(L)$, is

$$QF_0 - R, \quad Q = \prod_{i=2}^{m-1} \left(1 + L \frac{K_{i-1}}{K_i} \lambda\right),$$

$$F_0 = \lambda^3 + 11\lambda^2 + 31\lambda + 16, \quad R = 5\lambda^2 + 33\lambda + 16.$$

Every factor of Q has modulus at least $|1 + L\lambda|$ for $\Re \lambda = x \geq 0$. If $\nu \geq 2$, then

$$|Q|^2 \geq 1 + 2\nu Lx + \nu L^2 y^2 \geq 1 + Ax + \frac{5}{4}y^2, \quad A = 2\nu L.$$

On the certified domain $A \geq \sqrt{5\nu} \geq \sqrt{10} > 1/4$. With $z = y^2$ and $U = A - 1/4$, the exact source

$$E_{22} = [1 + (U + 1/4)x + (5/4)z] |F_0(x + i\sqrt{z})|^2 - |R(x + i\sqrt{z})|^2$$

has 22 nonzero monomials. All coefficient polynomials are nonnegative and equality is possible only at $\lambda = 0$. The coefficient $5/4$ is sharp for this certificate only: replacing it by B makes the $z = y^2$ coefficient at $x = 0$ equal to $256B - 320$. This does not identify an intrinsic dynamical boundary. Also

$$(QF_0 - R)'(0) = 16L \sum_{i=2}^{m-1} K_{i-1}/K_i - 2 > 0,$$

so the conservation zero is simple. If $\nu = 1$, putting $\gamma = 91L/90$ gives

$$\frac{QF_0 - R}{\lambda} = \gamma\lambda^3 + (1 + 11\gamma)\lambda^2 + (6 + 31\gamma)\lambda + (16\gamma - 2),$$

whose coefficients and Routh–Hurwitz difference $6 + 99\gamma + 325\gamma^2$ are positive because $\gamma > 1/8$.

The second harmonic is unchanged, while the zero harmonic changes by $w_0(L) = w_0^{\text{ref}} + \tau_m(L)\rho$. The cubic numerator is $N_m(L) = N_m^{\text{ref}} + \tau_m(L)S_m$. With $1/91 \leq \mathfrak{h}_m < 1/90$,

$$S_m = -\frac{4(1760850\mathfrak{h}_m - 10253)}{462105}, \quad -\frac{1}{10} < S_m < 0.$$

Since $C_m < 0$, exact clearing gives

$$N_m^{\text{ref}} - \frac{1}{100} \geq \frac{P_{\text{ref}}(\nu)}{D_{\text{ref}}(\nu)},$$

where

$$\begin{aligned} P_{\text{ref}} &= 3790502986637265684840\nu^5 - 974216530468600286489\nu^4 - 53103567440921218871\nu^3 \\ &\quad - 576386186827093561\nu^2 + 3649732858601219\nu + 55281268032918, \\ D_{\text{ref}} &= 715296950550(8\nu - 1)(90\nu + 1) \\ &\quad \times (589180301\nu^3 + 35065866\nu^2 + 629431\nu + 3306). \end{aligned}$$

$D_{\text{ref}} > 0$, and the shift $\nu = v + 1$ gives six positive coefficients for P_{ref} , proving $N_m^{\text{ref}} > 1/100$.

For the gauge, set

$$\begin{aligned} A_\tau &= 1494249120\mathfrak{h}_m L\nu^2 - 69786990\mathfrak{h}_m L\nu + 108738630L\nu^2 \\ &\quad + 1214388L\nu - 8521L - 125249670\nu^2 + 1031940\nu, \\ B_\tau &= 32760\mathfrak{h}_m L\nu + 32760L\nu^2 + 4L - 4095\nu. \end{aligned}$$

Then $\tau_m = -A_\tau/[15876(8\nu - 1)B_\tau]$ and $B_\tau = 4095\nu[8L(\nu + \mathfrak{h}_m) - 1] + 4L > 0$. The exact derivatives are negative on $L \geq 1/\sqrt{3\nu}$, so the maximum is at $(\mathfrak{h}_m, L) = (1/91, 1/\sqrt{3\nu})$. For $\nu \geq 2$ its endpoint sign is bounded by

$$P_{\text{up}}(\nu) = -\frac{189709065}{2}\nu^3 - 507201030\nu^2 - 935658\nu + 58481,$$

with

$$P'_{\text{up}} = -\frac{569127195}{2}\nu^2 - 1014402060\nu - 935658 < 0, \quad P_{\text{up}}(2) = -2789453215 < 0.$$

The case $\nu = 1$ uses the exact unreduced endpoint expression and $\sqrt{3} < 7/4$. Hence $\tau_m(L) < 1/20$. These bounds hold on the broader cubic range $L \geq 1/\sqrt{3\nu}$ and give $N_m(L) > 1/200$, hence $c_m(L) < 0$ throughout the certified interval.

The physical contrasts are

$$\chi_D = \frac{23}{63}91\nu L, \quad \chi_H = \frac{91\nu - 1}{91\nu L}.$$

Here χ_H measures equilibrium concentration-scale separation, not a universal measure of biological expense. Moreover

$$\chi_D(L)\chi_H(L) = \frac{23(91\nu - 1)}{63} = \frac{23(91m - 183)}{63} = \chi_D^{\text{unit}}(m).$$

Thus L reallocates a fixed product and reduces $\max(\chi_D, \chi_H)$, not the product. At L_0 ,

$$\chi_D = \frac{2093}{63}\kappa_\nu\sqrt{\nu}, \quad \chi_H = \frac{\sqrt{\nu}}{\kappa_\nu} \frac{91\nu - 1}{91\nu},$$

so both are $\Theta(\sqrt{m})$. The topology-specific necessary bound for stationary crossings implies any such realization must have $\max(\chi_D, \chi_H) > \sqrt{8\nu}$, so exponent $1/2$ is optimal among stationary-crossing realizations of $\hat{N}_m$. Constants remain open.

Failure tests. Verify the SCC exhaustion at $b = 2a$; the principal-minor derivative argument; the 22/35/77/84-term equality cases and the direct $\nu = 1$ cubic; the transformed physical diffusion; the fixed-mass gauge; and the comparison $N_m(L) > 1/200$. No finite regression substitutes for these all- m identities.